Spot Challenge - Marker Test Guide (Ubuntu)

This guide shows you how to set up the camera and test ArUco markers. **You do not need to write or change any code.** You only copy commands, paste them into the Terminal and press Enter.

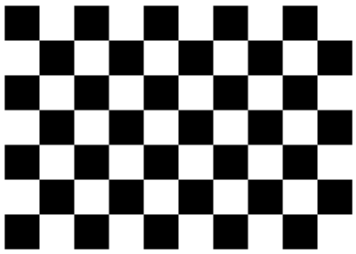
Part A uses the laptop webcam (practice). **Part B** uses the Tello's downward camera. **The drone never takes off in this guide.** The motors stay off.

What you need

1. Internet and a web browser (to download the kit)
2. A printer (A4 paper), a ruler, a measuring tape
3. Flat cardboard and glue or tape
4. A laptop with a webcam and internet (for step 2)
5. A charged Tello drone (the downward camera needs **Tello EDU / Tello TT**)

Important values (keep this page)

Item	Value	File to print
Chessboard	10 x 7 squares (9 x 6 inner corners) square size = 21 mm	print/chessboard_10x7_21mm_A4.pdf
ArUco markers	Dictionary (family): DICT_6X6_250 IDs: 0, 1, 2 black square = 150 mm	print/aruco_DICT_6X6_250_id0-2_150mm_A4.pdf









How to run a command

1. Copy one grey command box, paste it into the Terminal, press **Enter**.
2. A program opens a window. To close it: click on the window and press **Q**.
3. If a command shows red error text, look at "Problems and fixes" at the end.

Step 0 - Download the kit

1. Open this link in a web browser: **https://github.com/chissanupong-s/Spot_challenge_fidutial_marker**
2. Click the green **"<> Code"** button, then click **"Download ZIP"**.
3. Open your **Downloads** folder. Right-click `Spot_challenge_fidutial_marker-main.zip` and choose **"Extract Here"**.
4. Move the unzipped folder `Spot_challenge_fidutial_marker-main` to your Desktop (easy to find).
5. Open it. You should see the guides, a `print` folder and a `scripts` folder.

Important: Always work inside the unzipped folder, not inside the .zip file.

Step 1 - Print

1. Open `print/chessboard_10x7_21mm_A4.pdf` and `print/aruco_DICT_6X6_250_id0-2_150mm_A4.pdf`.
2. In the print window choose **"Actual size"** or **"100%"**. Do NOT choose "Fit to page".
3. **Measure with a ruler:** one chessboard square should be **21 mm**. The black ArUco square should be **150 mm**.
4. **Write down what you measured.** If it is a bit different (for example 20.5 mm), that is OK - you will type your measured number later.
5. Glue the chessboard **flat** on cardboard. It must not bend.
6. Cut the markers on the dashed line. **Keep the white border** around the black square. Do not fold them.

Note: Meters = millimeters / 1000. Example: 21 mm = 0.021, 20.5 mm = 0.0205, 150 mm = 0.15.

Step 2 - Install (only once)

Important: Do this while the laptop is connected to the normal internet (not the Tello Wi-Fi).

1. Open the folder `Spot_challenge_fidutial_marker-main` (from Step 0).
2. Right-click an empty area in the folder and choose **"Open in Terminal"**.
3. Paste these commands one by one (it asks for your password once):

```
sudo apt update
sudo apt install -y python3 python3-venv python3-pip
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

When you see **(.venv)** at the start of the line, you are ready.

Important: Every time you open a new Terminal: open it in the kit folder (as above) and run this first:

```
source .venv/bin/activate
```

Part A - Laptop webcam

Step 3 - Find the camera ID

A laptop can have more than one camera. Each camera has a number (ID): 0, 1, 2 ...

```
python -m scripts.aruco.check_cameras
```

1. A window shows the camera image with **CAMERA ID = 0**.
2. Press **N** to see the next camera. Press **Q** to close.
3. **Write down the ID** of the camera you want (usually 0 = built-in, 1 = USB camera).
4. In the next commands, replace `--camera-id 0` with your ID.

Step 4 - Take chessboard photos

```
python -m scripts.aruco.capture_calibration_images --camera webcam --camera-id
0
```

1. Hold the chessboard in front of the camera.
2. Green text **"BOARD FOUND"** = good: hold still and press **SPACE** to save a photo.
3. Red text = the camera cannot see the whole board. Move it until the text is green.
4. Take **20 to 30 photos**: near and far, tilted left / right / up / down, and in every corner of the picture.
5. Use good light. Avoid shiny reflections on the board.
6. Press **Q** when done.

Step 5 - Calibrate the webcam

```
python -m scripts.aruco.calibrate_camera --camera webcam --square-size 0.021
```

1. `--square-size` = **your measured square size in meters** (21 mm = 0.021).
2. Wait until it prints **"Calibration complete."**
3. Look at the line **"Mean reprojection error (pixels)"**: below 0.5 = great, below 1.0 = OK, above 1.0 = do Step 4 again (first delete the photos in `data/calib_images/webcam`).
4. The result is saved automatically in `assets/calibration/webcam`. You do not need to copy anything.

Step 6 - Test the printed markers

```
python -m scripts.aruco.webcam_distance --camera-id 0 --marker-length 0.15
```

1. `--marker-length` = **your measured black square in meters** (150 mm = 0.15).
2. Hold marker ID 0 in front of the camera. You see a green outline, colored axes and text like **"ID: 0 | Z: 0.503 m"**. Z = distance from the camera in meters.
3. Check it: put the marker **50 cm** and **100 cm** from the camera (use the tape). Z should be close (within about 2-3 cm). Write the numbers in the results table.
4. Try markers ID 1 and ID 2 too. Press **Q** to close.

Part B - Tello downward camera

Important: The drone does NOT take off in these steps. Hold it in your hand or put it on a box. Use a fully charged battery.

Step 7 - Connect and first test

1. Turn on the Tello and wait about 10 seconds.
2. Connect the laptop Wi-Fi to **TELLO-XXXXXX**. (No internet now - that is normal.)
3. Open the Terminal in the kit folder, run `source .venv/bin/activate`, then:

```
python -m scripts.tello.marker_test --camera down
```

1. A small grey picture appears - this is the camera under the drone.
2. Put a marker on the floor. Hold the drone 30-50 cm above it, camera facing down.
3. You see a green outline and text like **"ID:0 area:5200px"**.
4. "Calibration: NO" is normal now - distance comes after Step 9.
5. Press **V** to switch to the front camera and back. Press **Q** to close.

Important: If you see "Tello rejected the camera switch": this drone cannot use the downward camera (it needs Tello EDU / TT). Stop here and tell the dev team.

Step 8 - Chessboard photos with the downward camera

```
python -m scripts.aruco.capture_calibration_images --camera tello_down
```

1. Put the chessboard flat on the floor or a table, in good light.
2. Hold the drone above it, camera facing down, about 30-60 cm high.
3. This time **move and tilt the drone**, not the board.
4. Same rules as Step 4: green "BOARD FOUND" -> press **SPACE**. Take **20 to 30 photos**. Press **Q** when done.

Step 9 - Calibrate the downward camera

```
python -m scripts.aruco.calibrate_camera --camera tello_down --square-size
0.021
```

Check the "Mean reprojection error" the same way as Step 5. The result is saved in `assets/calibration/tello_down`.

Step 10 - Test markers with distance

```
python -m scripts.tello.marker_test --camera down --marker-length 0.15
```

1. Now the screen says **"Calibration: yes"** and shows **z** in meters.
2. Hold the drone **30 cm, 50 cm and 100 cm** above the marker (measure from the camera to the floor). Write down z each time.
3. Press **S** to save a picture at each height.

4. Try markers ID 1 and ID 2, and a marker near the edge of the picture.

Step 11 - Send the results to the dev team

Zip these and send them:

1. the folder `assets/calibration`
2. the folder `data/snapshots`
3. your results table (a photo of the paper is fine)

Camera	Marker ID	Real distance	Screen Z / z	OK?
Webcam	0	50 cm		
Webcam	0	100 cm		
Tello down	0	30 cm		
Tello down	0	50 cm		
Tello down	0	100 cm		

Problems and fixes

Problem	Fix
No module named 'cv2' or 'djitellopy'	You forgot to activate. Run: source .venv/bin/activate
No module named 'scripts'	The Terminal is not in the kit folder. Open it in the folder again.
Camera window is black / does not open	Close Zoom, Teams, browser camera tabs. Try another camera ID.
Text never turns green (Step 4 / 8)	More light, board flat, the WHOLE board in the picture, no reflections, hold still.
"Only N valid calibration images"	Take more photos (Step 4 / 8), then calibrate again.
"Calibration file not found"	Do the calibrate step (Step 5 / 9) first.
Marker is not detected	Show the whole marker with its white border. Keep it flat. Come closer. More light. Only the printed markers from this kit work.
WARNING: calibration ... does not fit	The calibration was made with another camera. Redo the photos + calibrate steps for this camera.
Tello: no picture / stuck	Is the Wi-Fi connected to TELLO-XXXXXX? Turn the drone off and on, wait 10 s, try again. If the firewall is on: sudo ufw allow 11111/udp